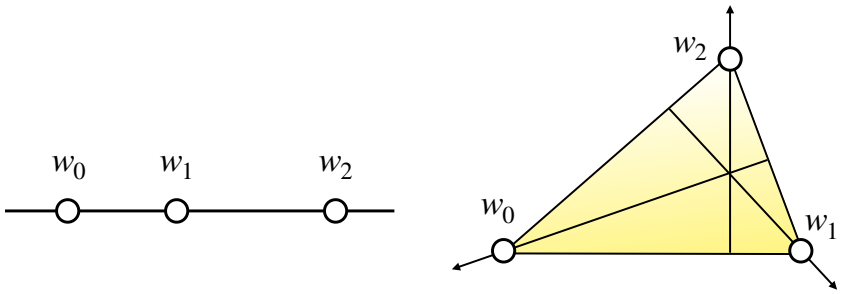

Figure 7: Illustration of the case $k = 3$ in the proof of Proposition 9.

(c) Let's say that w_1 is w_Z defined in Definition 2. The binary contrast $z_0 \Rightarrow z_1$ is subordinate to the binary feature for the attribute w_0. By the property of the causal inner product, $\bar{\ell}_{w_0}$ is orthogonal to the linear representation $\bar{\ell}_{z_0 \Rightarrow z_1} = \bar{\ell}_{z_1} - \bar{\ell}_{z_0}$ (by Corollary 10). Then, with the above result (b), we have $(\bar{\ell}_{w_1} - \bar{\ell}_{w_0})^\top (\bar{\ell}_{z_1} - \bar{\ell}_{z_0})$.

(d) By the above result (a), we have

$$\begin{cases} \|\bar{\ell}_{w_1} - \bar{\ell}_{w_0}\|_2^2 = \|\bar{\ell}_{w_1}\|_2^2 - \|\bar{\ell}_{w_0}\|_2^2 \\ \|\bar{\ell}_{w_2} - \bar{\ell}_{w_1}\|_2^2 = \|\bar{\ell}_{w_2}\|_2^2 - \|\bar{\ell}_{w_1}\|_2^2 \\ \|\bar{\ell}_{w_2} - \bar{\ell}_{w_0}\|_2^2 = \|\bar{\ell}_{w_2}\|_2^2 - \|\bar{\ell}_{w_0}\|_2^2. \end{cases} \tag{B.9}$$

Then,

$$\|\bar{\ell}_{w_1} - \bar{\ell}_{w_0}\|_2^2 + \|\bar{\ell}_{w_2} - \bar{\ell}_{w_1}\|_2^2 \tag{B.10}$$
$$= \|\bar{\ell}_{w_1}\|_2^2 - \|\bar{\ell}_{w_0}\|_2^2 + \|\bar{\ell}_{w_2}\|_2^2 - \|\bar{\ell}_{w_1}\|_2^2 \tag{B.11}$$
$$= \|\bar{\ell}_{w_2}\|_2^2 - \|\bar{\ell}_{w_0}\|_2^2 \tag{B.12}$$
$$= \|\bar{\ell}_{w_2} - \bar{\ell}_{w_0}\|_2^2. \tag{B.13}$$

Therefore, $\bar{\ell}_{w_1} - \bar{\ell}_{w_0}$ is orthogonal to $\bar{\ell}_{w_2} - \bar{\ell}_{w_1}$.

□

C Natural Categorical Concepts as Simplices

Polytopes are quite general objects. Definition 7 also includes representations of categorical variables that are semantically unnatural, e.g., $\{\texttt{dog}, \texttt{sandwich}, \texttt{running}\}$. We would like to make a more precise statement about the representation of "natural" concepts. One possible notion of a "natural" concept is one where the model can freely manipulate the output values. The next proposition shows such concepts have particularly simple structure:

Proposition 9 (Categorical Concepts are Represented as Simplices)**.** *Suppose that $\{w_0, \ldots, w_{k-1}\}$ is a collection of k mutually exclusive attributes such that for every joint distribution $Q(w_0, \ldots w_{k-1})$ there is some ℓ_i such that $\mathbb{P}(W = w_i \mid \ell_i) = Q(W = w_i)$ for every i. Then, the vector representations $\bar{\ell}_{w_0}, \ldots, \bar{\ell}_{w_{k-1}}$ form a $(k-1)$-simplex in the representation space. In this case, we take the simplex to be the representation of the categorical concept $W = \{w_0, \ldots, w_{k-1}\}$.*

Proof. If we can represent arbitrary joint distributions, this means, in particular, that we can change the probability of one attribute without changing the relative probability between a pair of other attributes. Consider the case where $k = 3$, as illustrated in Figure 7. If $\bar{\ell}_{w_0}, \bar{\ell}_{w_1}, \bar{\ell}_{w_2}$ are on a line, then there is no direction in that line (to change the value in the categorical concept) such that adding the direction can change the probability of w_2 without changing the relative probabilities between w_0 and w_1. However, if $\bar{\ell}_{w_0}, \bar{\ell}_{w_1}, \bar{\ell}_{w_2}$ are not on a line, they form a triangle. Then, there exists a line

that is toward $\bar{\ell}_{w_2}$ and perpendicular to the opposite side of the triangle. Now adding the direction $\tilde{\ell}$ can manipulate the probability of w_2 without changing the relative probabilities between w_0 and w_1. That is, for any $\alpha > 0$ and context embedding ℓ,

$$\begin{cases} \mathbb{P}(W = w_2 \mid \ell + \alpha\tilde{\ell}) > \mathbb{P}(W = w_2 \mid \ell), \text{ and} \\ \frac{\mathbb{P}(W=w_1 \mid \ell+\alpha\tilde{\ell})}{\mathbb{P}(W=w_0 \mid \ell+\alpha\tilde{\ell})} = \frac{\mathbb{P}(W=w_1 \mid \ell)}{\mathbb{P}(W=w_0 \mid \ell)}. \end{cases} \tag{C.1}$$

Therefore, the vectors $\bar{\ell}_{w_0}, \bar{\ell}_{w_1}, \bar{\ell}_{w_2}$ form a 2-simplex.

This argument extends immediately to higher k by induction. For each $i \in \{0, \dots, k-1\}$, there should exist a direction that is toward $\bar{\ell}_{w_i}$ and orthogonal to the opposite hyperplane ($(k-2)$-simplex) formed by the other $\bar{\ell}_{w_{i'}}$'s. Then, the vectors $\bar{\ell}_{w_0}, \dots, \bar{\ell}_{w_{k-1}}$ form a $(k-1)$-simplex. □

D Subspaces for Categorical Concepts

By Theorem 4, polytope representations have the following property:

Corollary 10 (Polytope Representations)**.** *Let $W = \{w_0, \dots, w_{k-1}\}$ be a categorical concept, and suppose there exist vector representations $\bar{\ell}_{w_i}$ for the binary features of each attribute w_i (their convex hull is the polytope representation of W). Let $\bar{L} = \left[\bar{\ell}_{w_1} - \bar{\ell}_{w_0}, \dots, \bar{\ell}_{w_{k-1}} - \bar{\ell}_{w_0}\right] \in \mathbb{R}^{d\times(k-1)}$ and let $\mathcal{C}(\bar{L})$ be its column space. Then, there exist some vectors $\tau_{w_i} \in \mathcal{C}(\bar{L})$, for each i, such that*

$$\begin{cases} \prod_{\bar{L}} g(y) = \tau_{w_i} & \text{if } y \in \mathcal{Y}(w_i) \\ \prod_{\bar{L}} g(y) = 0 & \text{if } y \notin \cup_{i=0}^{k-1}\mathcal{Y}(w_i), \end{cases} \tag{D.1}$$

where $\prod_{\bar{L}} = \bar{L}(\bar{L}^\top\bar{L})^\dagger\bar{L}^\top$ is the projection matrix onto $\mathcal{C}(\bar{L})$, and $(\bar{L}^\top\bar{L})^\dagger$ denotes a pseudo-inverse of the matrix.

Proof. By Theorem 4, for $i = 1, \dots, k-1$, we have

$$\prod_{\bar{L}} g(y) = \bar{L}(\bar{L}^\top\bar{L})^\dagger\bar{L}^\top g(y) = \bar{L}(\bar{L}^\top\bar{L})^\dagger \left[0, \dots, 0, b_{w_i}^2, 0, \dots, 0\right]^\top := \tau_{w_i} \tag{D.2}$$

for any $y \in \mathcal{Y}(w_i)$. Similarly, we have

$$\prod_{\bar{L}} g(y) = \bar{L}(\bar{L}^\top\bar{L})^\dagger\bar{L}^\top g(y) = \bar{L}(\bar{L}^\top\bar{L})^\dagger \left[-b_{w_0}^2, \dots, -b_{w_0}^2\right]^\top := \tau_{w_0} \tag{D.3}$$

for any $y \in \mathcal{Y}(w_0)$, while

$$\prod_{\bar{L}} g(y) = \bar{L}(\bar{L}^\top\bar{L})^\dagger\bar{L}^\top g(y) = \bar{L}(\bar{L}^\top\bar{L})^\dagger \left[0, \dots, 0\right]^\top = 0 \tag{D.4}$$

for any $y \notin \cup_{i=0}^{k-1}\mathcal{Y}(w_i)$. Therefore, we have eq. (D.1) with some constant vectors τ_{w_i}. □

In words, the unembeddings of tokens for the same attribute share the same projections on the subspace spanned by the differences between the vector representations. Therefore, adding any vector in the subspace $\mathcal{C}(\bar{L})$ to the context embedding changes the probability of the target concept $W = \{w_0, \dots, w_{k-1}\}$ *without* changing any other concept subordinate to or causally separable with the target concept. Finally, note that Corollary 10 implies Corollary 6.

E Experiment Details

We employ the `Gemma-2B` version of the Gemma model (Mesnard et al., 2024), which is accessible online via the `huggingface` library. Its two billion parameters are pre-trained on three trillion tokens. This model utilizes 256K tokens and 2,048 dimensions for the representation space.

We always use tokens that start with a space ('\u2581') in front of the word, as they are used for next-word generation with full meaning. Additionally, like WordNet data we use, we include plural

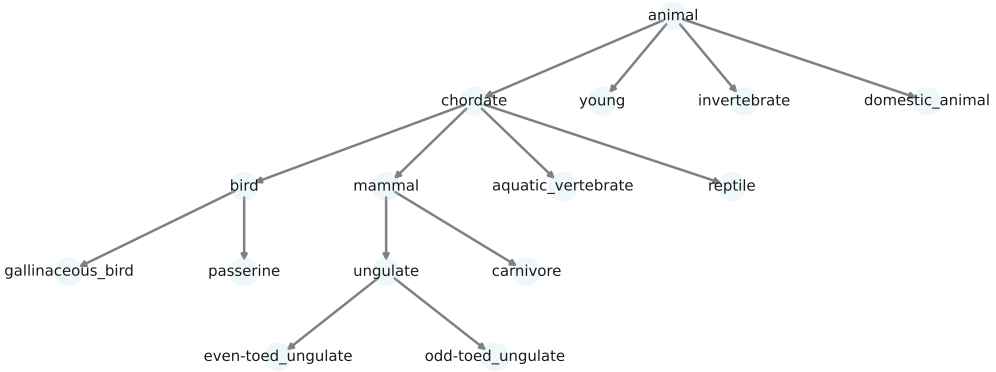

Figure 8: Subtree in WordNet noun hierarchy for descendants of `animal`.

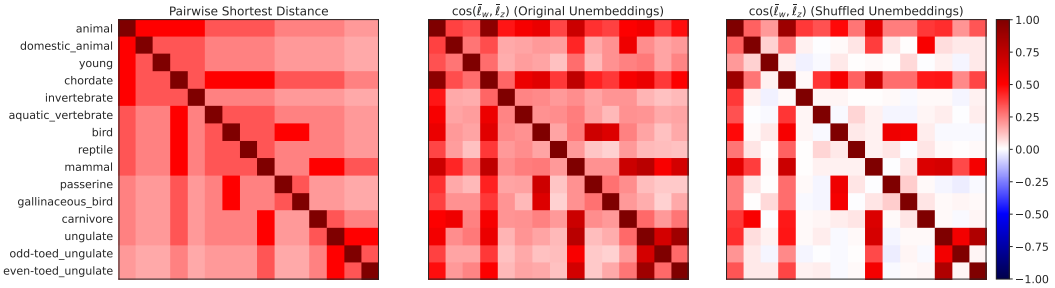

Figure 9: Zoomed-in Heatmaps of the subtree for `animal` in Figure 8.

forms, and both capital and lowercase versions of the words in $\mathcal{Y}(\texttt{animal})$ and $\mathcal{Y}(\texttt{plant})$ for visualization in Appendix A.

In the WordNet synset data, each content of the synset `mammal.n.01` indicates that "mammal" is a word, "n" denotes "noun," and "01" signifies the first meaning of the word. In the WordNet hierarchy, if a parent has only one child, we combine the two features into one. Additionally, since the WordNet hierarchy is not a perfect tree, a child can have more than one parent. We use one of the parents when computing the $\bar{\ell}_w - \bar{\ell}_{\text{parent of } w}$.

Lastly, we use the vector representations estimated by the non-split collection of tokens for Figure 2, Table 1, Figure 4, Figure 6, Figure 14, and Figure 18.

F Additional Results

Zooming in on a Subtree of Noun Hierarchy As it is difficult to understand the entire WordNet hierarchy at once from the heatmaps in Figure 4, we present a zoomed-in heatmap for the subtree (Figure 8) for the feature `animal` in Figure 9. The left heatmap displays the shortest distance between the nodes of the subtree in Figure 8. The middle heatmap shows that the cosine similarities between the vector representations $\bar{\ell}_w$ reflect the child-parent or sibling relationships. The right heatmap demonstrates that the vector representations estimated by shuffled unembeddings only reflect the set inclusion relationships, and other pairs have cosine similarities close to 0.

Additional Results on Hierarchical Orthogonality Analogous to Figure 5, we validate the statement (d) in Theorem 8. Figure 10 shows that a child-parent vector and a parent-grandparent vector for each feature estimated by the original unembedidngs (blue) are orthogonal. The random parent and grandparent vectors (orange) make the cosine similarity not close to 0, suggesting that the orthogonality is not merely a byproduct of the high-dimensional space.